Team 4
ONE WAY OUT (OWO)
Requirements:

1. The game shall be used with a computer with the proper requirements/specs to run the latest version of Windows 11.
    1.1. The computer shall have a processor, 1 gigahertz (GHz) or faster with 2 or more cores on a compatible 64-bit processor or System on a Chip (SoC).
    1.2. The computer shall have RAM: 4 gigabytes (GB).
    1.3. The computer shall have storage: 64 GB or larger storage device.
    1.4. The computer shall have system firmware of UEFI, Secure Boot capable.
    1.5. The computer shall have a TPM, Trusted Platform Module version 2.0.
    1.6. The computer shall have a graphics card compatible with DirectX 12 or later with a WDDM 2.0 driver.
    1.7. The computer shall have a display higher than 720p that is greater than 9” diagonally, 8 bits per color channel.
2. The game shall be controlled with a mouse and keyboard.
3. The game shall be connected to an audio device.
4. The game shall maintain a frame rate of at least 30 FPS on hardware meeting the specified minimum system requirements.
5. The game shall have a main menu which shall allow the player to start the game.
    5.1. The main menu shall be displayed once the player has pressed any key.
    5.2. The main menu shall display the title of the game.
    5.3. The main menu shall display the start button.
    5.4. The main menu shall display the settings button.
        5.4.1. The settings menu shall have a sound setting.
        5.4.2. The settings menu shall have a display/scaling setting.
    5.5. The main menu shall display the difficulty button.
        5.5.1. The difficulty button shall have an easy, medium, and hard difficulty that changes the experience of the game.

5.5.1.1. Easy difficulty shall have two puzzles in two of the rooms and two rooms giving the player the exam sheet (4 questions in total).
5.5.1.2. Medium difficulty shall have two puzzles in all four of the rooms (8 questions in total).
5.5.1.3. Hard difficulty shall have four puzzles in all four of the rooms (16 questions in total).

5.6. The main menu shall display the close button.
5.6.1. The close button shall exit the game.

6. The game shall have a UI that allows the player to be informed and interact within the game.
6.1. The UI shall allow the player to see their items.
6.1.1. The UI shall make a VFX (visual effect) for interacting with an item.
6.2. The UI shall allow the player to look at their objective.
6.2.1. The UI shall update the player’s objective when the player completes the previous one.
6.3. The UI shall allow the player to use settings.
6.3.1. The player shall be able to pause the game at any time during gameplay by pressing the ESC, escape key.
6.4. The UI shall allow the player to see the timer.
6.5. The UI shall allow the player to see the professor's alertness meter that indicates the number of incorrect questions made.
6.6. The UI shall allow the player to see their custom cursor icon.
6.6.1. The cursor shall change when hovering over an interactable item.
6.6.2. The cursor shall change when hovering over an interactable door.
6.6.3. The cursor shall change when hovering over an interactable puzzle.

7. The game shall allow the player to interact with objects in the environment by clicking on them with the mouse.
7.1. The game shall allow the player to interact with doors that allow the player to progress through the house.
7.2. The game shall allow the player to interact with puzzles that prompt the player with a math problem.

8. The game shall allow the player to do math puzzles that progress the game.
    8.1. The puzzles shall include genres of math that include multiplication, division, fractions, and geometry.
    8.2. The puzzles shall no longer be interactable once the player solves it correctly.
    8.3. The puzzles shall be located in different rooms throughout the house.
9. The player shall be placed in a gameplay loop that demands all required puzzles to be completed.
10. The game shall determine whether the player's submitted answer is correct or incorrect.
    10.1. The game shall punish the player by increasing the professor's alertness meter.
        10.1.1. The alertness meter is proportional to the number of incorrect tries the player makes.
        10.1.2. The game difficulty shall determine how many total times a player can make an incorrect answer.
            10.1.2.1. Easy difficulty shall have 15 attempts.
            10.1.2.2. Medium difficulty shall have 8 attempts.
            10.1.2.3. Hard difficulty shall have 3 attempts.
    10.2. The game shall trigger a losing condition when the professor's alertness meter reaches its maximum.
11. The game shall start a timer during gameplay that will cause the player to lose when the time reaches zero.
    11.1. The timer shall be set to 5 minutes
    11.2. The timer shall only start once the player finishes the tutorial at the start of the game.
        11.2.1. The tutorial will prompt a simulation of how the gameplay loop operates.
        11.2.2. The tutorial begins once the player enters the basement.
        11.2.3. The tutorial ends once the player leaves the basement.
    11.3. The timer shall pause when the player pauses the game.
    11.4. The game shall display a losing screen when the countdown timer reaches zero or the professor's alertness reaches its maximum.

11.4.1. The game shall display a cutscene of the school where the player is in the principal's office.
11.4.2. The game shall display an “EXPELLED” stamp indicating the player loses.
11.4.3. The game shall allow options for how the player wants to continue.
11.4.4. The game shall prompt the player with a restart button.
11.4.5. The game shall prompt the player with a main menu button.

12. The game intro shall display an introductory cutscene explaining the story before gameplay begins.
12.1. The game shall provide a tutorial in the basement that introduces the player to the gameplay mechanics.
12.1.1. The tutorial shall give examples of the interactions the player can do.
12.1.1.1. Interaction shall include a math puzzle.
12.1.1.2. Interaction shall include unlocking doors.
12.1.1.3. Interaction shall include moving through doors.
12.2. The game shall reward the player with a basement key after completing the math puzzle.

13. The game shall allow the player to exit the basement after obtaining the basement key.
14. The game shall start once the player has exited the basement into the hallway
15. The game shall prevent the player from exiting the house until all required exam sheets have been collected.
15.1.1. The game shall display dialogue informing the player that they can’t leave if they attempt to exit without all the required exam sheets.
15.1.2. The game shall allow the player to unlock the exit after collecting all required exam sheets.
15.2. The game shall allow the player to navigate between the hallway and available rooms by clicking on doors or other navigation options.
15.2.1. The game screen shall have a fade-to-black transition for each room.

15.2.2. The game shall display a back arrow that allows the player to return to the previous area.

15.3. The game shall reward the player with an exam sheet after completing all the required puzzles in a room.

15.4. The game shall prevent duplicate puzzle rooms and questions from appearing more than once during a single playthrough.

15.5. The game shall allow the player to leave a puzzle room before completing all its puzzles.

16. The game shall display an ending screen based on the outcome of the game.

16.1. The game shall display a victory screen when the player completes the required objectives and escapes.

16.1.1. The game shall pan out to a bird's-eye view of the professor's house when the player escapes.

16.1.2. The game shall display a victory message.

16.1.3. The game shall display the amount of time it took to complete.

17. The game shall have ambience audio and music during gameplay.

18. The game shall have music during the menu screen.

19. The game shall have music playing during the victory screen.

20. The game shall have music playing during the losing screen.